"6D_1 Two Scale Factors" is the Case where the two curled dimensions form a torus, T^2, and 3-space and curled space have individual scale factors.

Notation and Setup:

Starting Metric: $g_{00}=c^2$, $g_{11}=-a[t]^2\,B[r]$, $g_{22}=-r^2\,a[t]^2\,C[r]$, $g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]$, $g_{44}=-b[t]^2\,p_4^2$, $g_{55}=-b[t]^2\,p_5^2$, a(t) and b(t) are independent and not constant

Corresponding Line Element

$ds^2 = c^2\,dt^2 - a[t]^2\,B[r]\,dr^2 - r^2\,a[t]^2\,C[r]\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]\,d\phi^2 - b[t]^2\,p_4^2\,d\psi_4^2 - b[t]^2\,p_5^2\,d\psi_5^2$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, **$A'=\partial_r A(t,r)$** and **$\dot{A}=\partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) independent scale factors a and b, (2) $\mathcal{A}(t) = c^2$, and (3) C left free, which gives the FG-GRW form of the solution; setting C=1 gives the GRW form.

a: GRW Equations

We begin with the assumptions:

a(t) and b(t) are independent and non-constant,

$A(t,r) = c^2$,

B(r) is a free parameter whose formula we seek,

C(r) is a free parameter,

and $\mathcal{D}_4[r] = \mathcal{D}_5[r] = 1$ (which make the torus Ricci flat).

The initial Lambda field equations are:

$$R_{00}: \quad \frac{3\,\ddot{a}}{a\,c^2} + \frac{2\,\ddot{b}}{c^2\,b[t]} = \Lambda$$

$$R_{01}: \quad 0 = 0$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{2\,\dot{a}\,\dot{b}}{a\,c^2\,b[t]} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} - \frac{C''}{a^2\,C\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{2\,\dot{a}\,\dot{b}}{a\,c^2\,b[t]} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{33}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{2\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} + \frac{2\,\dot{a}\,\dot{b}}{a\,c^2\,b[t]} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,c^2\,b[t]} + \frac{\dot{b}^2}{c^2\,b[t]^2} + \frac{\ddot{b}}{c^2\,b[t]} = \Lambda$$

$$R_{55}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,c^2\,b[t]} + \frac{\dot{b}^2}{c^2\,b[t]^2} + \frac{\ddot{b}}{c^2\,b[t]} = \Lambda$$

R_{33} and R_{55} are redundant, the same as R_{22} and R_{44}, respectively.

R_{01} is an identity because $A(t,r) = c^2$.

R_{00} is the first of the equations we are seeking.

$R_{11}-2R_{22}$ can be rearranged to have only r terms on the left and only t-terms on the right, so both sides equal a constant that we call $-2\,c^2\,k$:

$$R_{11}-2R_{22}: \quad -\frac{2\,c^2}{r^2\,C} + \frac{2\,c^2}{r^2\,B} + \frac{2\,c^2\,C'}{r\,C\,B} + \frac{c^2\,(C')^2}{2\,C^2\,B} = -a^2\,c^2\,\Lambda + 2\,\dot{a}^2 + a\,\ddot{a} + \frac{2\,a\,\dot{a}\,\dot{b}}{b[t]} = -2\,c^2\,k$$

$$\Rightarrow \quad (1) \quad \frac{2\,c^2}{r^2\,B} + \frac{2\,c^2\,C'}{r\,C\,B} + \frac{c^2\,(C')^2}{2\,C^2\,B} = -2\,c^2\,k + \frac{2\,c^2}{r^2\,C}$$

$$\text{and} \quad (2) \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{2\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda - \frac{2\,c^2\,k}{a[t]^2}$$

Solving (1) for B yields the same result as in 4D:

$$B(r) = \frac{(2\,C + r\,C')^2}{4\,C\left(1 - k\,r^2\,C\right)}.$$

Plugging this in, the equations become:

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} + \frac{2\,\ddot{b}}{b[t]} = c^2\,\Lambda$$

$$R_{11}: \quad \frac{2\,c^2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{2\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda$$

$$R_{22}: \quad \frac{2\,c^2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{2\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{\dot{b}^2}{b[t]^2} + \frac{\ddot{b}}{b[t]} = c^2\,\Lambda$$

$$R_{55}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{\dot{b}^2}{b[t]^2} + \frac{\ddot{b}}{b[t]} = c^2\,\Lambda$$

R_{55} and R_{22} have become redundant, the same as R_{44} and R_{11}, respectively.

We are left with three ordinary differential equations:

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} + \frac{2\,\ddot{b}}{b[t]} = c^2\,\Lambda$$

$$R_{11}: \quad \frac{2\,c^2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{2\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{\dot{b}^2}{b[t]^2} + \frac{\ddot{b}}{b[t]} = c^2\,\Lambda$$

As a check, Case 6D_1 was developed for only one scale factor, a.
The solution required k=0, so setting k=0 and b=a, we get the same three equations found in the 6D_1 notebook's Section e, the Vacuum Friedmann Equations:

$$R_{00}: \quad \frac{5\,\ddot{a}}{a} = c^2\,\Lambda$$

$$R_{11}: \quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = c^2\,\Lambda$$

$$R_{44}: \quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = c^2\,\Lambda$$

Moreover, the Friedmann continuity equation is derived as an algebraic combination of these equations:

$$-\frac{1}{2}R_{00} + \frac{3}{2}R_{11} + R_{44}: \quad \frac{3\,c^2\,k}{a^2} + \frac{3\,\dot{a}^2}{a^2} + \frac{6\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{\dot{b}^2}{b[t]^2} = 2\,c^2\,\Lambda$$